Session 10: Strings — Notes & Code (C Language)

Mastering String Functions, Operations, and Memory Safety in C

October 11, 2026

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1 Topic Notes

1.1 What is a String?

- In C, a string is a **one-dimensional character array** terminated by a null character (`'\0'`).
- Declared as: `char str[100];`
- The null character signals to functions where the string memory ends.

1.2 Common String Functions

C provides string manipulation utilities in the `<string.h>` standard library header.

Function	Purpose	Return Value
`strlen(str)`	Returns length of string (excluding `'\0'`)	`size_t` (integer)
`strcpy(dest, src)`	Copies source string into destination buffer	Address of `dest`
`strcmp(str1, str2)`	Compares two strings lexicographically	`0` if equal, `<0` or `>0`

Table 1: Key Functions in `<string.h>`

1.3 Key Technical Points

- `strlen()` counts active characters only, excluding the terminating null character.
- The destination array must have **sufficient capacity** prior to invoking `strcpy()` to prevent buffer overflows.
- `strcmp()` returns `0` when identical, a negative value when `str1 < str2`, or a positive value when `str1 > str2`.
- **Important:** Never use the == operator to compare string contents in C; it evaluates memory addresses rather than character values. Use `strcmp()` instead.

2 Task 1: Length of a String using `strlen()`

2.1 Code

```c
#include <stdio.h>
#include <string.h>

int main() {
    char songTitle[] = "Tum Hi Ho";

    int length = strlen(songTitle);

    printf("Song Title: %s\n", songTitle);
    printf("Length of the string: %d\n", length);

    return 0;
```

```
}
```

Listing 1: Calculating String Length in C

2.2 Output

```
Song Title: Tum Hi Ho
Length of the string: 9
```

3 Task 2: Compare Two Usernames using `strcmp()`

3.1 Code

```c
#include <stdio.h>
#include <string.h>

int main() {
    char user1[50], user2[50];

    printf("Enter first username: ");
    scanf("%s", user1);

    printf("Enter second username: ");
    scanf("%s", user2);

    if (strcmp(user1, user2) == 0) {
        printf("The usernames are the SAME.\n");
    } else {
        printf("The usernames are DIFFERENT.\n");
    }

    return 0;
}
```

Listing 2: Comparing String Contents with strcmp()

3.2 Output

```
Enter first username: alex
Enter second username: alex
The usernames are the SAME.
```

4 Task 3: Copy a String using `strcpy()`

4.1 Code

```c
#include <stdio.h>
#include <string.h>

int main() {
    // Declare enough space for destination string
    char shoppingApp[20];

    strcpy(shoppingApp, "Flipkart");

    printf("Shopping App: %s\n", shoppingApp);

    return 0;
}
```

Listing 3: Safe String Copy Operation

4.2 Output

```
Shopping App: Flipkart
```

Note: `shoppingApp` is initialized with a size of 20 elements, ensuring adequate room for the characters of `"Flipkart"` along with the null terminator `'\0'`.

5 Task 4: Create Username from Full Name

5.1 Notes

- Extracts up to the first 5 characters from a full name to form a short username.
- Uses `strncpy()` for bounded string copying.
- **Critical Step:** Manually appends the null terminator `'\0'` at index 5 when the input contains 5 or more characters, as `strncpy()` does not guarantee null-termination when maximum count is reached.

5.2 Code

```c
#include <stdio.h>
#include <string.h>

int main() {
    char fullName[100];
    char username[10];   // 5 characters + '\0'

    printf("Enter your full name: ");
    scanf(" %[^\n]", fullName);   // reads line including spaces

    if (strlen(fullName) >= 5) {
        // Copy only first 5 characters
        strncpy(username, fullName, 5);
        username[5] = '\0';   // manually add null terminator
    } else {
        // Name shorter than 5 characters -> use full name
        strcpy(username, fullName);
    }

    printf("Generated Username: %s\n", username);

    return 0;
}
```

Listing 4: Generating Bounded Username from Full Name

5.3 Output 1

```
Enter your full name: Rohan Sharma
Generated Username: Rohan
```

5.4 Output 2

```
Enter your full name: Amy
Generated Username: Amy
```

6 Quick Revision Summary

Task / Concept	Primary Utility	Key Takeaway
String Length	`strlen()`	Measures total characters excluding `'\0'`.
String Comparison	`strcmp()`	Returns 0 on exact match; do not use `==`.
String Copy	`strcpy()`	Destination buffer must be sufficiently sized.
Bounded Copy	`strncpy()`	Explicitly append `'\0'` after copying.

Table 2: String Operations Summary

Core Takeaway: C strings are manual character buffers without implicit runtime checks. Always ensure destination buffers have capacity for content plus the terminating `'\0'` byte.